

Sanjana Venkatesh

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Education

University of Southern California, USA
Master of Science, Analytics (Data Science)

August 2023 - May 2025
GPA: 3.8

BMS College of Engineering, India
Bachelor of Engineering, Information Science and Engineering

August 2017 - July 2021
GPA: 3.71

Technical Skills

- **Programming Languages** - Python, SQL (MySQL, PostgreSQL, Snowflake, BigQuery), Java, Scala, C++, SAP
- **Big Data and Cloud Technologies** - Apache Spark, Hadoop (HDFS, Hive, Impala), Databricks, Google Cloud Platform (BigQuery, Dataflow, Dataproc, Pub/Sub), AWS (S3, Glue, Lambda, Redshift)
- **ETL and Data Pipelines** - Apache Airflow, dbt, Kafka, Informatica, Talend, Google Cloud Composer
- **Data Warehousing and Storage** - Snowflake, BigQuery, Amazon Redshift, Hive, Delta Lake
- **Data Processing and Analytics** - Pandas, NumPy, PySpark, Pytorch, Looker, Tableau, ThoughtSpot, Power BI
- **DevOps and Automation** - Git, Docker, CI/CD (GitHub Action), API Development (REST), Version Control

Experience

Data Engineer Intern

Los Angeles, CA

SAG-AFTRA Health | SAG-Producers Pension

May 2024 - August 2024

- Engineered an automated **metadata management system** in Python, dynamically updating **ThoughtSpot** worksheet descriptions from Snowflake, enhancing data integrity and reducing manual intervention by 57%
- Created a **REST API** integration to automate data ingestion and transformation from ThoughtSpot API into **Snowflake**, optimizing ETL workflows and reducing data retrieval time by 42%
- Enhanced and optimized dimensional data models using **Star** and **Snowflake schemas** in **BigQuery** and **Redshift**, improving query performance by 28% and supporting scalable analytics solutions
- Built a comprehensive **User Adoption Dashboard** leveraging **ThoughtSpot** and **Tableau**, delivering actionable insights on user engagement, adoption trends, and dashboard performance to drive data-informed decision-making

Data Engineer

Bangalore, India

Quantiphi Analytics

February 2021 - July 2023

- Optimized and migrated on-prem **Hadoop** jobs to **Google Cloud Dataproc**, cutting costs by 41%, and implemented **CI/CD** pipelines for secure deployment while automating job orchestration with **Cloud Composer**, saving 8+ hours of manual work.
- Designed and managed **ETL pipelines** using **Apache Airflow** and **Python**, developing custom DAGs to automate data ingestion and transformation, cutting manual processing time by 55%, improving data integrity and accessibility for business teams.
- Enhanced cluster performance using **Spark UI**, fine-tuning **Dataproc Cluster (EFM)** with SSD balancing and preemptive workers, ensuring resilience to input data skew and workload management.
- Led a team of 8 engineers to migrate 1,500+ tables from **Cloudera** to **GCP**, ensuring 99% data accuracy and compliance with business reporting standards using **Data Validation Tool (DVT)**.
- Built and automated dynamic **Power BI** and **Tableau** dashboards to visualize trading characteristics, benchmarks, and performance KPIs, enhancing financial decision-making and improving efficiency by 25%.
- Executed a seamless data migration from **Amazon RDS PostgreSQL** to **GCP AlloyDB**, implementing optimizations that ensured zero data loss or downtime.

Projects

Retail Data Analytics | *Tableau, Apache Airflow, dbt, BigQuery, GCS, Python*

- Orchestrated an **ETL pipeline** using Apache Airflow DAGs, automating data ingestion from **GCS** to **BigQuery**
- Constructed **dbt SQL models** for data transformation, optimizing query performance and ensuring high data integrity
- Launched interactive Tableau dashboards, reducing manual reporting by 48% and enabling real-time insights into revenue trends, top-selling products, and market performance

Real-Time Stock Data Analytics | *Kafka, Spark, InfluxDB, Grafana, PostgreSQL, Python*

- Implemented a real-time stock data pipeline using **Kafka** and **Spark Streaming**, processing and storing stock data in **InfluxDB**
- Developed interactive dashboards using **Grafana** to visualize stock trends and support real-time decision-making
- Automated stock data ingestion and transformation with Python producer scripts, and managed static stock data in **PostgreSQL**

CHD Risk Analysis & Prediction | *Vertex AI, Python, SQL, Tableau*

- Engineered an automated ML pipeline using **Vertex AI**, identifying Coronary Heart Disease (CHD) with **Random Forest**
- Crafted interactive dashboards in Tableau, reducing reporting time by 40% and enhancing visualization of risk factors
- Streamlined data processing workflows for 500K+ patient records, increasing data accuracy by 20%